

PKCERT AI & Software Development Internship

Task 09 – Support Vector Machines (SVM) & k-Nearest Neighbors (kNN)

Objective:

Develop practical skills in implementing and evaluating SVM and kNN classification algorithms.

Total Marks: 100

Part A – Dataset Selection & Preparation (15 Marks)

1. Select a public classification dataset.
2. Describe the dataset, features and target.
3. Preprocess the dataset (missing values, encoding, scaling).
4. Split the dataset into training and testing sets.

Part B – Support Vector Machine (35 Marks)

1. Build an SVM model.
2. Choose an appropriate kernel.
3. Evaluate using Accuracy, Precision, Recall, F1-Score and Confusion Matrix.
4. Explain SVM, its advantages, limitations and applications.

Part C – k-Nearest Neighbors (35 Marks)

1. Build a kNN model.
2. Choose an appropriate K value.
3. Evaluate using Accuracy, Precision, Recall, F1-Score and Confusion Matrix.
4. Explain kNN, its advantages, limitations and applications.

Part D – Comparative Analysis (15 Marks)

1. Compare SVM and kNN based on performance and recommend the better model for the selected dataset.

Assessment Area	Marks
Dataset Preparation	15
Support Vector Machine	35

k-Nearest Neighbors	35
Comparative Analysis	15